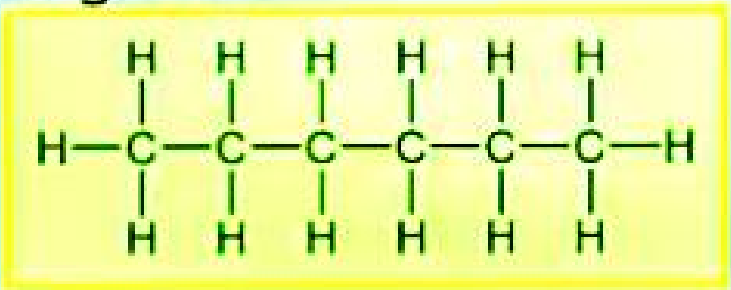
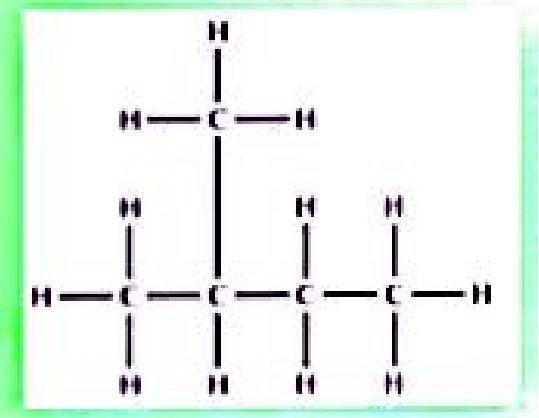
structural formula

- unambiguously shows how the atoms are bonded together



- can use ***condensed structural formulas***
  - bonds are omitted, **repeated groups put together**, **side chains put in brackets**
    - $CH_3CH_2CH_2CH_2CH_2CH_3$
      - or even $CH_3(CH_2)_4CH_3$
    - $CH_3CH(CH_3)CH_3$



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Preparation of alkanes

1- Hydrogenation of unsaturated hydrocarbon:

$$H_2C{=}CH_2 \xrightarrow[200,\ 300]{Ni\ or\ Pd\ or\ Pt\ /\ H_2} H_3C{-}CH_3$$

2- Reduction of alkyl halides

a- Hydrolysis of Grignard reagent

$$CH_3CH_2Br + Mg^{2+} \xrightarrow{Dry\ ether} CH_3CH_2MgBr$$

Grignard reagent

$$CH_3CH_2MgBr \xrightarrow{H_3O^+} CH_3CH_3 + Mg(OH)Br$$

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b- Reduction by metal and acid

$$CH_3CH_2CH_2Br + Zn^{2+} \xrightarrow{H^+} CH_3CH_2CH_3 + ZnBr$$

$$CH_3CH_2CH_2CH_2Br \xrightarrow[2)\ H_3O^+]{1)\ LiAlH_4\ /\ Ether} CH_3CH_2CH_2CH_3$$

c- From coupling with metal

$$2\ H_3C{-}Br + 2\ Na \longrightarrow H_3C{-}CH_3 + 2\ NaBr$$

d- Using cuprate method (Gilman reagent)

$$(CH_3CH_2)_2CuLi + CH_3Br \longrightarrow CH_3CH_2CH_3$$

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Reaction of alkanes

1- Halogenation

$$RH + X_2 \xrightarrow[\text{or UV light}]{\text{Heat}} RX + HX$$

$X_2 = Cl_2$ or Br_2

Alkyl Halide

Free radical mechanism

$$CH_4 + Cl_2 + \text{energy} \rightarrow CH_3Cl + HCl$$

$$Cl_2 \text{ or } Br_2 \xrightarrow{h\nu / \Delta} 2\,Cl^{\cdot}$$

$$\dot{Cl} + CH_4 \longrightarrow HCl + CH_3^{\cdot}$$

$$\dot{CH_3} + \dot{Cl} \longrightarrow CH_3Cl$$

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if we have one type of carbon

$$CH_4 + Cl_2 + \text{energy} \longrightarrow CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4 + HCl$$

If we have different type of carbon

Selectivity in halogenation reactions

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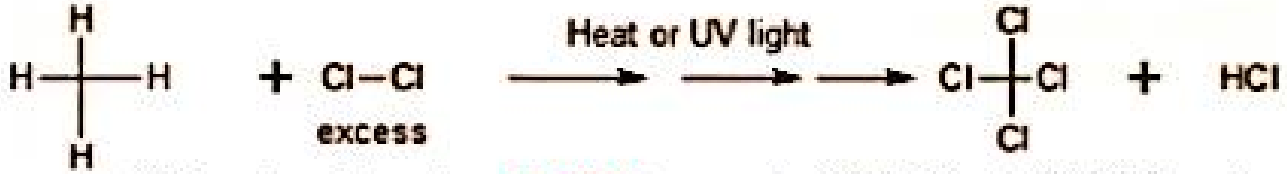
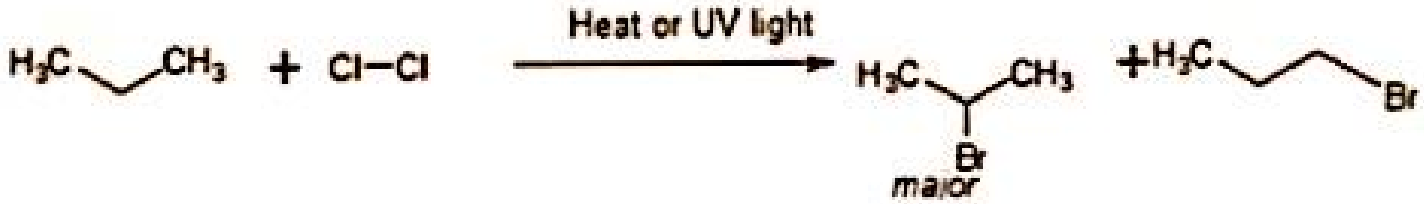
if we have one type of carbon



$$CH_4 + Cl_2 + \text{energy} \longrightarrow CH_3Cl + CH_2Cl_2 + CHCl_3 + CCl_4 + HCl$$

If we have different type of carbon

Selectivity in halogenation reactions



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2- combustion of alkanes

$$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O + \underline{213\ Kcal\backslash mol}$$

$$C_2H_6 + 7/2\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O + \underline{373\ Kcal\backslash mol}$$

160 kcal\mol for each methylene group

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